Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3371-WE01

Title: (BEng/MEng)

Electrical Engineering 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Section A

Question 1

a) A separately excited DC Motor has an armature resistance of 0.5 Ω which is connected at a 240 V RMS, 50 Hz single phase AC source through a single phase full wave thyristor converter. The motor is operating at full load. The field is kept constant by maintaining a constant field voltage across the field windings.

i. Draw the output waveform of the single phase full wave thyristor converter supplying the armature winding and show that the average DC output voltage of the converter (i.e. the motor supply voltage) is determined by the following expression:

$$V_d = 2 \times V_s \frac{\sqrt{2}}{\pi} \times \cos \alpha$$

In which Vs is the RMS of the AC input single phase voltage and α is the thyristor converter firing angle

[50%]

ii. With the firing angle set to zero (i.e. α=0), the motor is energised and runs at no-load speed of 2500 RPM. Calculate the motor constant (in V/RPM) and the back EMF in this case. Assume armature current is negligible in this case.

[20%]

b) For the DC motor in part (a), with the firing angle still set at zero, the motor is loaded and now runs at 2325 RPM. Determine the back EMF, power delivered to the load from the motor and the power drawn by the motor from the converter in this case.

[30%]

continued

Question 2

A two pole, 50 Hz, 400 V, three-phase star-connected induction motor is driven by a three-phase PWM controlled Voltage Source Inverter (VSI). The motor is under scaler control (V/Hz control) and it is initially started with rated voltage and rated frequency (400 V/50 Hz). Parameters of the per-phase equivalent circuit of this motor are given in Table Q2.

Table Q2 – Induction Motor Per-phase Equivalent Circuit Parameters
(All values referred to the stator)

Parameter	**Stator Winding**	**Rotor (referred to the stator)**	**Magnetising Branch**
Resistance	$0.12\ \Omega$	$0.10\ \Omega$	-
Reactance	$0.25\ \Omega$	$0.25\ \Omega$	$10\ \Omega$

Resistive component of the magnetising branch is ignored.

a) Sketch a block diagram of a three-phase scalar control system of a three-phase induction motor and clearly label all components. Briefly explain how this control system can be used for dynamic electromagnetic braking, and name one industrial application.

[30%]

b) This motor drives an industrial fan. The torque required for the fan varies as the square of the angular speed and is given by:

$$T_m = 1.36 \times 10^{-3}(\omega_m)^2$$

For 5 % slip, determine the speed and torque of the fan when the motor is supplied by the rated voltage and rated frequency.

[20%]

c) The motor drive setting is now changed to 200 V/25 Hz. Sketch the torque-speed characteristic of the motor for both 400 Hz/50 Hz, and 200 V/25 Hz and qualitatively describe the operating point of the motor for both settings.

[20%]

d) Use the simplified per-phase equivalent circuit of the three-phase induction motor, and calculate the line current at starting point $(S = 1)$, and under full load condition $(S = 5\ \%)$, when the PWM VSI is set at the rated voltage and rated frequency of the motor.

[30%]

continued

Section B

Question 3

Consider a 50 Hz three-phase medium length transmission line with a series impedance of

$$Z = R + jX = 11 + j47\ \Omega.$$

When the phase voltage at the receiving end of the line is $127\angle 0$ kV and the three-phase load at the receiving end is $30\,MW$ with a power factor of 0.8 lagging, the sending end voltage is $V_S = 224.72\angle 1.446°$ kV.

a) Find the current at the receiving end (I_R) in this case.

[20%]

b) Find the generalised line constants (A, B, C, D).

[20%]

c) Check your answers for A, B, and I_R, using the equation $V_S = AV_R + BI_R$ and show that using the calculated values for A, B, and I_R, the sending end voltage (V_S) will be calculated correctly.

[10%]

d) Find the magnitude of the receiving end voltage if the load at the receiving end is 30 MW with a power factor of 0.8 lagging, and the line-line sending end voltage is 220 kV.

[50%]

Question 4

a) Consider a short transmission line, where we want to increase the active power transfer capability of the line by installing a FACTS device at either the sending or the receiving end, or at the middle of the line. Describe at least three solutions in full detail and compare those solutions in terms of cost and performance, as a function of location and type of the device.

[80%]

b) Explain why there is one and only one slack bus when modelling a power system for power flow studies.

[20%]

END